The Edge Effect

Where Should You Focus to Raise NPS?

The customer problem with the most ratings from 0 to 6 may not be the problem that raises NPS fastest.

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ABSTRACT

Net Promoter Score (NPS) begins with customer ratings from **0 to 10** and converts them into a company score from **−100 to +100**. When several customer problems compete for attention, which one can raise NPS fastest? We built a computer program that ran a Monte Carlo experiment, using repeated randomized what-if testing, across **12,000 fictional companies**, each with 20,000 customers: **240 million ratings in total**. It compared the largest and lowest-scoring problems with problems containing customers closest to changing NPS group. It also varied starting NPS, score spread, success chances, and how one to six equal-sized improvement steps were divided. The formula points to a clear short-term rule; the simulation tests when it holds and where it fails. The result identifies immediate score opportunities, not necessarily the problem causing the greatest customer harm or the best long-term experience.

THE SHORT ANSWER

For each customer problem, first count the linked survey responses scored exactly 6 or 8.

A rating of 6 is one point from leaving the detractor group, while an 8 is one point from entering the promoter group. If the problems require similar time and cost and have a similar chance of raising customer ratings, the problem with the largest combined count offers the most immediate opportunities to raise NPS.

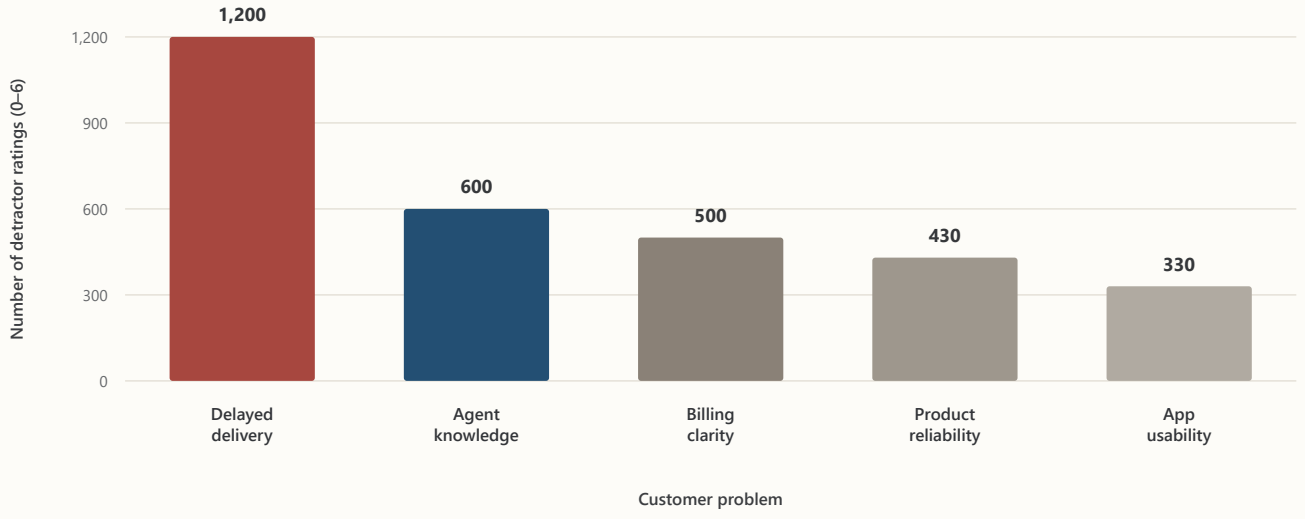
1	**Count the opportunities:** add the 6s and 8s for every customer problem.	**2**	**Check the real world:** which problem is affordable and likely to improve?	**3**	**Track the lowest ratings separately:** do not ignore serious problems simply because a small improvement may not change NPS immediately.

What "problem" means here: it is the main issue linked to an NPS response, for example, delayed delivery or support-agent knowledge. A company might identify it from a written comment or feedback tag. The model compares problem categories, not individual customers.

BEGIN WITH A FAMILIAR DECISION

The largest customer problem looks like the obvious priority. Then we see where its ratings fall.

Imagine that a company uses each customer's comment to link the NPS response to one main problem. Customers scoring from 0 to 6 are called **detractors**. The company ranks its problem categories by the number of detractor responses and sees this:



Illustrative example. The vertical axis counts detractor ratings from 0 to 6; the horizontal axis lists the reported customer problems. These are invented numbers used only to explain the question.

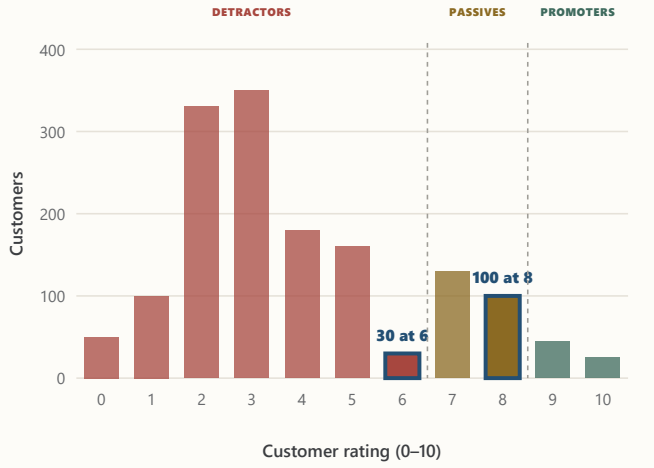
THE TEMPTING DECISION

Delayed delivery has twice as many detractors as agent knowledge. Improve it first, right?

The totals hide the part that changes the decision: where those customers sit on the 0–10 rating scale.

#1 BY DETRACTOR COUNT

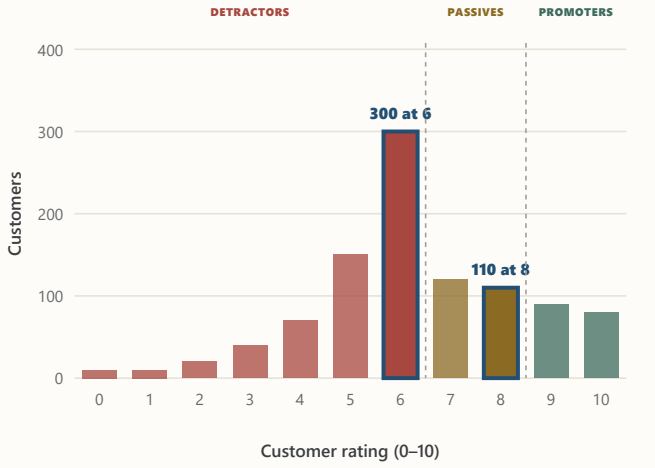
Delayed delivery — 1,200 detractors



Most ratings are near 2 or 3. Adding one modeled rating point changes those scores, but few customers leave the detractor group.

#2 BY DETRACTOR COUNT

Agent knowledge — 600 detractors



This problem has half as many detractors, yet many ratings are already at 6. A one-point rise moves those customers into the passive group.

WHAT THE TOTALS HIDE

The larger problem has 130 ratings of exactly 6 or 8. The smaller problem has 410.

A 6 is one point from leaving the detractor group, and an 8 is one point from entering the promoter group. The smaller problem therefore has more than **three times as many** immediate NPS-changing opportunities, despite having half as many detractors. At the 6-to-7 boundary alone, it has 300 ratings of 6 compared with 30, a tenfold difference. Delayed delivery may still cause greater harm. The narrower question is: *which problem has more customers who could change NPS group after the same amount of work?*

The total number of responses shows how widespread a problem is. The rating distribution shows how many customers are one point from changing NPS group.

HOW THE SIMULATION WORKS

The companies are invented. The experiment creates different companies and score patterns; it does not imitate a particular industry.

One improvement step represents one equal unit of limited resources. Those resources can mean investment, staff time, or operational effort. One step attempts to raise the rating of each successfully affected customer in one problem by one point. Two steps attempt a two-point increase, up to 10. A step is a comparison unit, not an NPS point or a fixed amount of money.

The main test assumes a 35% success chance. Each affected customer has a 35% chance of gaining one rating point. Sensitivity tests then allow this chance to differ by problem.

Six rules used to compare different customer problems

Each response has one main problem. Examples include delayed delivery, product reliability, and agent knowledge. Real problems can overlap.

One to six steps are separate resource scenarios. The program tests what to do with one, two, three, and up to six available resource units. With three units, it compares all three in one problem, 2 + 1 across two problems, and 1 + 1 + 1 across three.

The first comparison assumes equal-sized work. A real decision should also compare time, money, operational difficulty, risk, and customer harm.

1 The only two one-point rating changes that raise NPS

Customers answer the recommendation question from 0 to 10. Scores 0–6 are **detractors**, 7–8 are **passives**, and 9–10 are **promoters**. Passives remain part of the total number of responses, but they are neither added nor subtracted in the formula:

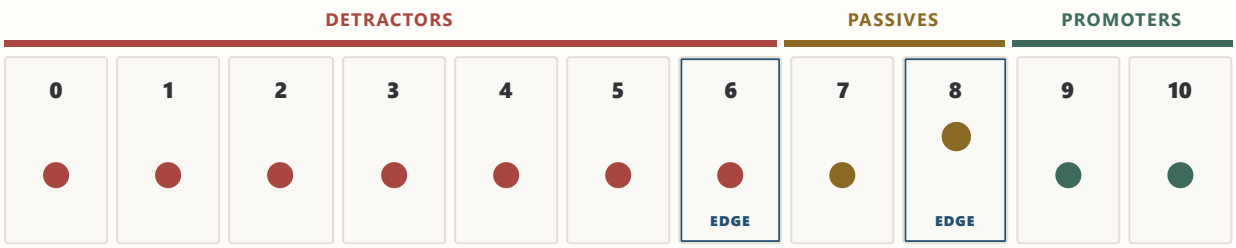
NPS = percentage of promoters – percentage of detractors

NPS ranges from −100 to +100. For example, if 50 of 100 customers are promoters and 20 are detractors, NPS is 50 − 20 = 30. Moving one of those 100 responses from 6 to 7 or from 8 to 9 raises NPS from 30 to 31. In a larger survey, one customer changes the total by less.

We will call 6 and 8 the two **edges** because they sit immediately below an NPS boundary. A one-point increase changes NPS only at these two places.

NPS MECHANICS

A one-point rating increase changes NPS only at two boundaries



6 → 7 removes one detractor

8 → 9 adds one promoter

Which boundary is better? Neither. A 6 → 7 change and an 8 → 9 change raise NPS by the same amount. If cost and difficulty are similar, choose the boundary with more customers likely to rise by one point.

Figure 1. An improvement from 6 to 7 removes one detractor; an improvement from 8 to 9 adds one promoter. Other one-point increases raise the reported rating but do not immediately change NPS.

This suggests a simple rule: choose the problem with the most ratings of 6 and 8. Next, we test whether that rule still works across many different companies.

2 From one company to 12,000

The opening company might be unusual, so we repeated the decision across 12,000 score patterns and starting NPS values. Each fictional company had 20,000 ratings across 12 generic problem categories. With one step, the program chose one problem. With several steps, it tested how to divide them.

12,000	×	20,000	=	240 million
FICTIONAL COMPANIES		CUSTOMERS EACH		SYNTHETIC RATINGS GENERATED

STEP 1

Create a fictional company

Assign 20,000 ratings to 12 problem categories and calculate a starting NPS.

STEP 2

Apply each selection rule

Choose a problem using rules such as largest category, most detractors, or most ratings at 6 and 8. Separately test every way to divide one to six steps.

STEP 3

Compare the outcomes

Calculate the NPS increase and the share of customers scoring 0–2 whose modeled rating rises.

Decision 1 · One equal-sized intervention	Decision 2 · Several equal-sized interventions
With one step, only 6 → 7 and 8 → 9 change NPS. Under equal work and success chance, counting 6s and 8s gives the best immediate choice.	With several steps, all ratings matter. Two steps can move 5 → 7 or 7 → 9, so the program tests every allocation.

We repeated the test four ways: every problem had the same success chance; each had a different random chance; tightly grouped ratings had a higher chance; or widely scattered ratings had a higher chance.

STRENGTH OF THE RESULT

What does the formula settle, and what needs simulation?

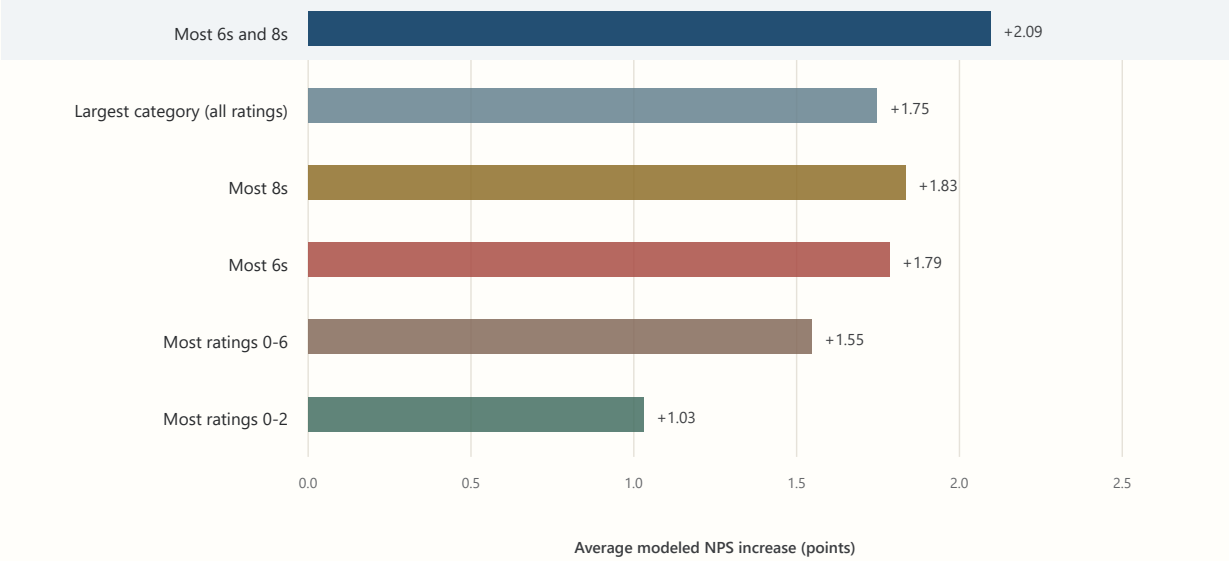
The clear case	With one step, equal work, and equal success chances, the problem with the most 6s and 8s has the largest immediate NPS opportunity.
Why simulate?	Simulation tests what happens when those conditions change or when several steps are available.
The critical limit	Simulation shows how the score may react. A real-world test is still needed to show whether an action improves the customer experience.

3 What the simulations found

FINDING 1

Which selection rule produced the largest modeled NPS increase?

Best simple rule: count ratings of 6 and 8 together. Every problem required the same amount of work, and each affected customer had a 35% chance of gaining one rating point.



How to read: each row is a rule for choosing one customer problem. A longer bar means a larger average NPS increase. The size-only rule chooses the category linked to the most responses, regardless of how its ratings are distributed.

Figure 2. With equal work and a 35% success chance, counting 6s and 8s produced an average gain of 2.09 NPS points. Choosing the largest category by all responses produced 1.75. The direct count led the size-only rule in all four success settings.

It led when every problem had equal conditions. The harder test assigned different success chances to different problems.

SENSITIVITY TEST

Did the direct count still lead when assumed success chances changed?

For each success-chance assumption, the program compared 15 rules for choosing one problem. Each row shows the 6-and-8 count beside the best-performing one of the other 14 rules. The numbers are average simulated NPS increases.

How success chances differ	Gain from counting 6s and 8s	Best alternative and gain	Which performed better?
Same rate for every problem	**2.095**	Average-near-edge rule: 1.968	**Count leads**
Different random rate for each problem	**2.191**	Average-near-edge rule: 2.047	**Count leads**
Tightly grouped ratings have a higher chance	**2.144**	Tightly grouped, average-near-edge rule: 2.084	**Count leads**
Widely scattered ratings have a higher chance	**2.052**	Average-near-edge rule: 2.065	**Near-tie: alternative +0.013**

What the alternatives mean: average-near-edge rules favor large problems whose average rating is close to 6 or 8. Three versions used narrow, medium, or broad definitions of "close." Only the final row favored an alternative rule. Comparing the rules within each fictional company, the paired 95% interval was 0.002 to 0.024 after allowing for all 14 alternatives. This is a consistent near-tie inside the model, not a meaningful practical advantage.

Starting NPS changes the size of the opportunity. The count rule gained 2.10 points from NPS 40 to under 70, 1.37 from 70 to under 85, and 0.74 at 85 or higher. Higher-scoring fictional companies had fewer ratings at 6 or 8.

FINDING 2

Can score spread or a category average replace the actual count?

We tested two tempting shortcuts. Neither should replace the actual 6-and-8 count without evidence.

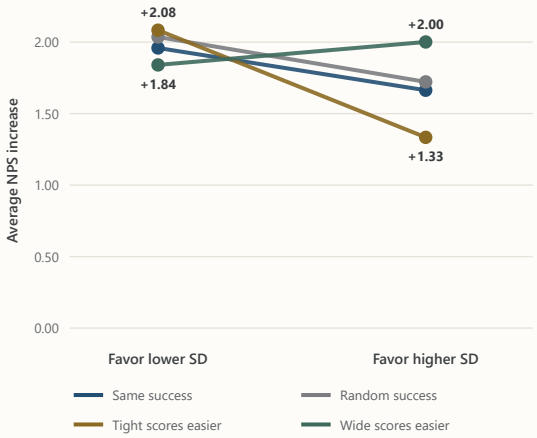
Shortcut 1 · Score spread: choose by how tightly ratings cluster. **Shortcut 2 · Category average:** choose a large problem whose average is near 6 or 8. We tested both against counting actual 6s and 8s.

Low standard deviation (low SD)	High standard deviation (high SD)
Ratings sit close together. If their average is near 6 or 8, many individual ratings may be nearby.	Ratings are widely spread. Even if their average is near 6 or 8, many ratings may be far away.

What SD tested here: both rules favored large categories whose average was near 6 or 8, then favored either lower or higher SD. Within these edge-near rules, the low-SD version led in three of four settings. The high-SD version led only when high-SD categories were given a higher success chance.

Tightly grouped or widely scattered scores?

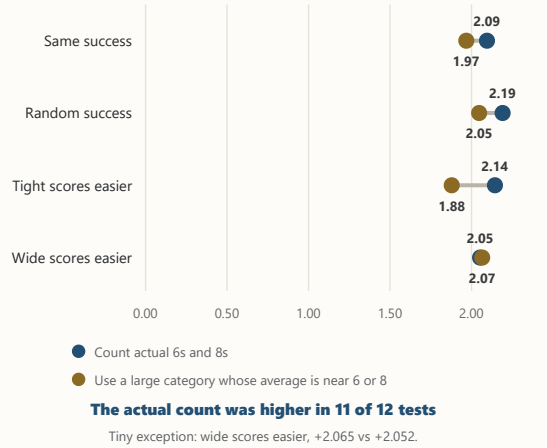
If two problem categories are already near an NPS edge, should the company favor ratings that sit close together or far apart?



Message: among large, edge-near categories, lower SD led in three settings. It is not a general measure of ease.

Count actual 6s and 8s, or use the category average?

Could the company skip the individual count and choose a large problem category whose average rating sits near 6 or 8?



Message: the actual count is safer because an average can hide very different score patterns. Farther right means a larger NPS increase.

Why an average can mislead: imagine two problems that both average 6. In one, every customer gave a 6. In the other, half gave 2 and half gave 10. Only the first problem has many customers one rating point from an NPS boundary.

Figure 3. Left: among size-weighted, edge-near rules, the lower-SD version led in three of four settings. Right: counting actual 6s and 8s led in 11 of 12 comparisons with average-based rules.

FINDING 3

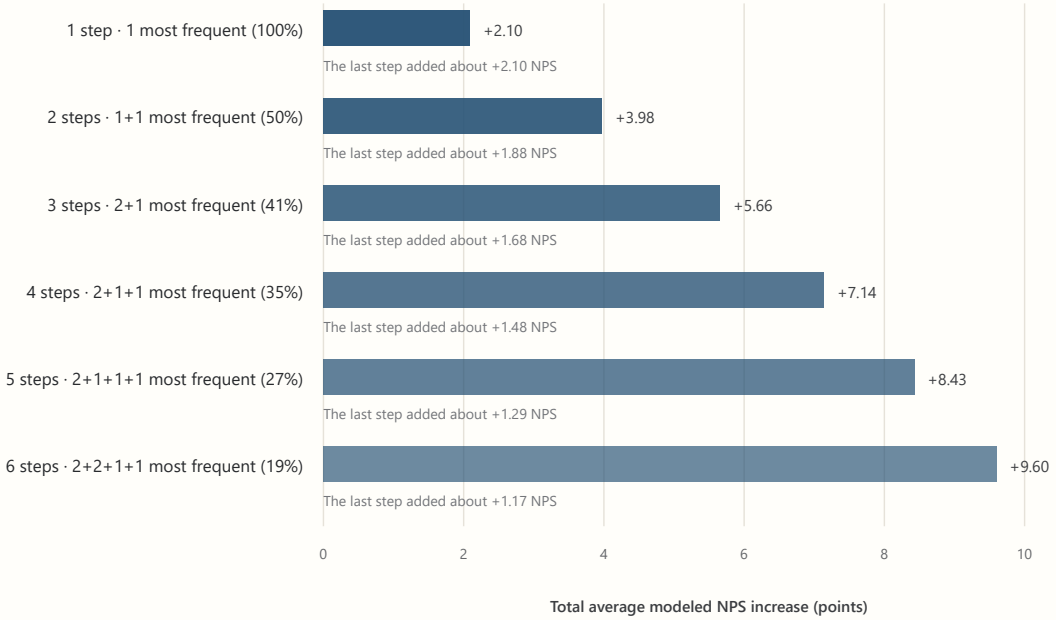
When more improvement work is available, should it go to one problem or several?

What is being divided? With three equal-sized steps, the program compares 3 on one problem, 2 + 1 across two problems, and 1 + 1 + 1 across three. Each bar shows a separate total from one to six steps.

What this chart assumes: in the three views with different success chances, the program is told each problem's chance of improvement. A real company would estimate those chances from past results or a small test.

No single split won everywhere.

Under the same success chance for every problem, the program found the best split separately for every fictional company. The most frequent six-step split won in only 19% of companies.



How to read: each bar shows the average best result. Its label shows the most common split, not a universal recommendation. "2 + 1" means two steps on one problem and one on another.

▼ **Compare all four success-chance settings at a glance**

How success chances are assigned	Most common allocation with 3 steps	Best plan uses several problems (3 steps)	Most common allocation with 6 steps	Best plan uses several problems (6 steps)
the same success chance for every problem	2 + 1 (41%)	70%	2 + 2 + 1 + 1 (19%)	97%
different known success chances by problem	2 + 1 (42%)	68%	2 + 2 + 1 + 1 (20%)	96%
tightly grouped ratings given higher success chances	1 + 1 + 1 (38%)	72%	2 + 1 + 1 + 1 + 1 (25%)	98%
widely scattered ratings given higher success chances	2 + 1 (44%)	65%	3 + 2 + 1 (19%)	95%

Figure 4. No split was best for every fictional company.

But who benefits from that allocation? Next, we check whether it reaches customers giving the lowest ratings.

FINDING 4

With 3 improvement steps, does maximizing NPS reach customers currently rating 0–2?

This comparison uses equal work and a 35% success chance for every problem.

PLAN A · LARGEST EXPECTED NPS INCREASE	PLAN B · VERY-LOW-SCORE SAFEGUARD
+5.66 modeled NPS points	**+4.88 modeled NPS points**
About **9%** of customers currently scoring 0–2 are modeled to gain rating points.	About **17%** of customers currently scoring 0–2 are modeled to gain rating points.

The tradeoff: Plan B reaches more customers having very bad experiences, while Plan A produces the larger NPS increase. Plan B reserves one step for the problem with the most ratings from 0 to 2, then uses the remaining steps to maximize NPS.

4 Discussion: feedback on this paper

The simulation results provide a starting point for a broader discussion about how companies use NPS. The following feedback, received on this paper and approved for inclusion, brings academic and managerial perspectives to that discussion.

Fred Reichheld

Creator of NPS; Bain & Company

> "A mathematically effective strategy may increase NPS, but leaders should focus on learning from root causes and improvement suggestions rather than targeting the score itself. Earned growth provides a stronger basis for accountability."

Bill Price

Former worldwide customer service vice president, Amazon; CEO, Intendra AI

> "I agree that counting sixes and eights for each customer problem is a useful approach, while recognizing that problems differ considerably in the time and expense required to resolve them. I also applaud the attention to deep detractors: improving their experience can reduce their likelihood of leaving, even when their ratings remain within the detractor category."

Sven Bähre

Professor of Quantitative Marketing and Market Research, Ulm University of Applied Sciences

> "I see this as an accessible examination of a relevant and underexplored managerial question. The comparison between maximizing NPS and reaching customers with very low ratings is particularly valuable because it reveals an important managerial tradeoff. When costs and success probabilities are equal, edge prioritization follows directly from the NPS formula. The simulation adds most when intervention effectiveness varies and resources must be distributed across issues. Managerial use should distinguish modeled NPS gains from customer and business outcomes, evaluate expected impact relative to the resources required, consider whether moving a 6 to 7 and an 8 to 9 have different economic value, and recognize that a large synthetic sample demonstrates the model's behavior rather than real-world validity."

Peter Fader

Frances and Pei-Yuan Chia Professor of Marketing, The Wharton School, University of Pennsylvania

> "Despite my deep admiration for the origin story behind NPS and the impact that it has had (and continues to have) for senior executives, I have concerns about some of the underlying measurement aspects. For instance, NPS is limited by arbitrary cutoffs and potential bias/measurement errors. Additionally, behavioral data often can provide a more rigorous basis for understanding customers."

Roland T. Rust

Distinguished University Professor and David Bruce Smith Chair in Marketing, University of Maryland; Executive Director, Center for Excellence in Service

> "Although NPS is widely used in business, it is not considered a leading academic measurement approach or demonstrably superior to available alternatives. Research seeking a broader academic contribution should look beyond NPS alone."

Comments on this paper were received between August and November 2026. The wording reproduced here was approved by each contributor.

Their feedback puts the practical rules that follow in context: the simulation identifies opportunities for immediate NPS movement, while customer benefit and business value require evidence beyond the score.

5 How to use this result in a real company

1 **Count scores 6 and 8 for each problem.** The largest combined count is the first candidate, not an automatic decision.	**2** **Generally prefer low SD near an edge.** Among large categories averaging near 6 or 8, the lower-SD rule led in three settings. The higher-SD rule led only when those categories were made easier to change.
3 **Prefer spreading resources when several units are available.** As the available resources increased, dividing them across several problems, usually in one-unit and two-unit portions, beat placing everything into one problem most often.	**4** **Do not assume size or severity raises NPS fastest.** The largest category, the category with the most detractors, and the category with the most ratings from 0 to 2 gained 1.75, 1.55, and 1.03 points, respectively, compared with 2.09 from counting 6s and 8s.

ONE-SENTENCE CONCLUSION

For immediate modeled NPS, begin with actual counts of 6s and 8s, generally prefer low-SD categories among large edge-near categories, and spread limited resources across several problems rather than concentrating everything in one.

> **Why NPS should not stand alone.** This note studies how the NPS formula reacts to score changes. Use it alongside complaints, retention, customer value, and direct evidence of improvement.

▼ References and reproducibility

1. Reichheld, F. F. (2003). The One Number You Need to Grow.
2. Keiningham, T. L. et al. (2007). A Longitudinal Examination of Net Promoter and Firm Revenue Growth.
3. Bain & Company. Measuring Your Net Promoter Score.
4. Fisher, N. I. & Kordupleski, R. E. (2019). Good and Bad Market Research: A Critical Review of Net Promoter Score.
5. Baehre, S. et al. (2022). Customer Mindset Metrics: A Systematic Evaluation of the Net Promoter Score (NPS) vs. Alternative Calculation Methods.

Companion files: **Ref. simulate.py** (Monte Carlo source code), **Ref. results.json** (simulation output), and **Ref. README.md** (reproduction guide). These files are supplied with the paper.

AI contribution

The research concept, experimental design, simulation logic, and testing decisions were developed independently of AI assistance. AI tools were used to support simulation code development, language refinement, and visual design and editing.